Department of Mathematics, IIT Madras
MA2101 Linear Algebra via Matrices
Assignment 2

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1. Check whether the given vectors are linearly independent:
   (a) $(1, 2, 6)$, $(-1, 3, 4)$, $(-1, -4, 2)$ in $\mathbb{R}^3$ **Answer:** Yes
   (b) $(1, 0, 2, 1)$, $(1, 3, 2, 1)$, $(4, 1, 2, 2)$ in $\mathbb{C}^4$ **Answer:** Yes
2. Suppose that u, v, w are linearly independent in $\mathbb{C}^5$. Are the following vectors linearly independent?
   (a) $u + v,\ v + w,\ w + u$
   (b) $u - v,\ v - w,\ w - u$
   (c) $u,\ v + \alpha w,\ w$, where α is a nonzero real number.
   (d) $u + v,\ v + 2w,\ u + \alpha v,\ v - \alpha w$, where α is a complex number.
3. Give three linearly dependent vectors in $\mathbb{R}^2$ such that none of the three is a scalar multiple of another.
4. Suppose S is a finite set of vectors, and some $v \in S$ is not a linear combination of other vectors in S. Is S linearly independent?

   Answer: No. Consider the set $S = \{(1,0,0,0), (0,1,0,0), (1,1,0,0), (0,0,1,0), (0,0,0,1)\}$. Here, $(0,0,1,0), (0,0,0,1)$ are not a linear combination of other vectors in S.

5. Prove that nonzero vectors $v_1, \ldots, v_m \in \mathbb{F}^n$ are linearly dependent iff there exists $k > 1$ such that the vectors $v_1, \ldots, v_{k-1}$ are linearly independent and v_k is a linear combination of $v_1, \ldots, v_{k-1}$.
6. Using elementary row operations determine whether the given vectors are linearly dependent or independent in each of the following cases, by taking the given vectors as rows of a matrix.
   (a) $[1, 2, 3]$, $[4, 5, 6]$, $[7, 8, 9]$
   (b) $[1, 0, -1, 2, -3]$, $[-2, 1, 2, 4, -1]$, $[3, 0, -1, 1, 1]$, $[2, 1, 1, -1, -2]$
   (c) $[1, 0, -1, 2, -3]$, $[-2, 1, 2, 4, -1]$, $[3, 0, -1, 1, 1]$, $[2, -1, 0, -7, 3]$
   (d) $[1, i, -1, 1 - i]$, $[i, -1, -i, 1 + i]$, $[2, 0, 1, i]$, $[1 + i, 1 - i, -1, -i]$
7. Solve Exercise 1 by forming a matrix with its columns as the transposes of the given vectors. If the vectors turn out to be linearly dependent, then express one of them as a linear combination of the others.
8. Let $A = [u_1, u_2, u_3, u_4, u_5] \in \mathbb{F}^{4\times 5}$. In each of the following cases determine the RREF of A.
   (a) u_1, u_2, u_3 are linearly independent; $u_4 = u_1 + 2u_2$, $u_5 = u_1 - u_2 + 2u_3$.
   (b) u_1, u_2, u_4 are linearly independent; $u_3 = u_1 + 2u_2$, $u_5 = u_1 - u_2 + 2u_4$.
   (c) u_1, u_3, u_5 are linearly independent; $u_2 = u_1 + 2u_3$, $u_4 = u_1 - u_3 + u_5$.
9. Answer the following questions with justification:
   (a) Is every subset of a linearly independent set linearly independent? **Answer:** Yes
   (b) Is every subset of a linearly dependent set linearly dependent? **Answer:** No
   (c) Is every superset of a linearly independent set linearly independent? **Answer:** No
   (d) Is every superset of a linearly dependent set linearly dependent? **Answer:** Yes
   (e) Is union of two linearly independent sets linearly independent? **Answer:** No
   (f) Is union of two linearly dependent sets linearly dependent? **Answer:** Yes
   (g) Is intersection of two linearly independent sets linearly independent? **Answer:** Yes
   (h) Is intersection of two linearly dependent sets linearly dependent? **Answer:** No
10. Determine rank r of the following matrices. Find out the r linearly independent rows, and also the r linearly independent columns of the matrix. And then express $4 - r$ rows as linear combinations of those r rows, and $5 - r$ columns as linear combinations of those r columns.

(a) $\begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 2 & 1 & 0 & 1 \\ 3 & 0 & 3 & 0 & 3 \\ 1 & 1 & 0 & 2 & 0 \\ 5 & 0 & 5 & 0 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 5 & 0 & 4 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 8 & 0 & 7 & 0 \end{bmatrix}$

11. Let $A \in \mathbb{F}^{n\times n}$. Prove that A is invertible iff $\text{rank}(A) = n$ iff $\det(A) \neq 0$.
12. Let $A \in \mathbb{F}^{m\times n}$. Let $P \in \mathbb{F}^{m\times m}$. When does the RREF of PA coincide with the RREF of A?
13. Let $u, v \in \mathbb{F}^{n\times 1}$. What is the rank of the matrix uv^t? **Answer:** 1
14. Show that if $A \in \mathbb{F}^{n\times n}$ has rank 1, then there exist $u, v \in \mathbb{F}^{n\times 1}$ such that $A = uv^t$.
15. Determine whether the following linear systems are consistent or not.
    (a) $2x_1 - x_2 - x_3 = 2,\ x_1 + x_3 - 4x_4 = 1,\ x_2 - x_3 - 4x_4 = 4$
    (b) $2x_1 - x_2 - x_3 = 2,\ x_1 + x_3 - 4x_4 = 1,\ 3x_1 - x_2 - 4x_4 = 4$
16. For each of the following augmented matrices, determine whether the corresponding linear system has (i) no solutions, (ii) a unique solution, (iii) infinitely many solutions.
    (a) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (b) $\left[\begin{array}{ccc|c} 1 & -2 & 2 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$ (c) $\left[\begin{array}{ccc|c} 1 & -1 & 3 & 8 \\ 0 & 1 & 2 & 7 \\ 0 & 0 & 1 & 2 \end{array}\right]$
17. In the following cases, what do you conclude about the number of solutions of the linear system $Ax = b$?
    (a) $A \in \mathbb{F}^{5\times 3}$ and $b = A_{\star 1} + A_{\star 2} = A_{\star 2} - A_{\star 3}$
    (b) $A \in \mathbb{F}^{3\times 4}$ and $b = A_{\star 1} + A_{\star 2} + A_{\star 3} + A_{\star 4}$
    (c) $A \in \mathbb{F}^{3\times 3}$ and $b = 3A_{\star 1} + 2A_{\star 2} + A_{\star 3} = 0$

    Answer: (a) If $rank(A) = 3$, then unique solution and if $rank(A) < 3$, infinitely many solutions.
    (b) Infinitely many solutions.
    (c) If $rank(A) = 3$, then only trivial solution and if $rank(A) < 3$, infinitely many solutions.
18. Let A be a $7 \times n$ matrix of rank r, and let $b \in \mathbb{F}^{7\times 1}$. For each of the following cases, determine, if possible, the number of solutions of the linear system $Ax = b$.
    (a) $n = 8,\ r = 5$ (b) $n = 8,\ r = 7$ (c) $n = 6,\ r = 6$ (d) $n = 6,\ r = 5$
19. Using Gauss-Jordan elimination, and also Gaussian elimination, solve the following linear systems:
    (a) $3w + 2x + 2y - z = 2,\ 2x + 3y + 4z = -2,\ y - 6z = 6$
    (b) $w + 4x + y + 3z = 1,\ 2x + y + 3z = 0,\ w + 3x + y + 2z = 1, 2x + y + 6z = 0$
    (c) $w - x + y - z = 1,\ w + x - y - z = 1,\ w - x - y + z = 2, 4w - 2x - 2y = 1$
20. For each of the following augmented matrices, determine the solution set of the corresponding linear system:
    (a) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$ (b) $\left[\begin{array}{ccc|c} 1 & 4 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right]$ (c) $\left[\begin{array}{ccc|c} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$
21. Let $B = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}$, and let $b = \begin{bmatrix} 0 \\ 5 \\ 3 \\ 4 \end{bmatrix}$.

    Suppose A is a matrix such that $A_{\star 1} = [2, 1, -3, 2]^t$, $A_{\star 2} = [-1, 2, 3, 1]^t$, B is the RREF of A, and $Aw = b$. Determine the following:
    (a) $\text{Sol}(A, 0)$ (b) $\text{Sol}(A, b)$ (c) A

    Answer: (a) $\alpha_3 \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_5 \begin{bmatrix} 1 \\ 1 \\ 0 \\ -5 \\ 1 \end{bmatrix} : \alpha_3, \alpha_5 \in \mathbb{F}$, (b) $\begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_5 \begin{bmatrix} 1 \\ 1 \\ 0 \\ -5 \\ 1 \end{bmatrix} : \alpha_3, \alpha_5 \in \mathbb{F}$

22. Let A be a matrix with $\mathrm{Sol}(A, 0) = \{\alpha[1\ 0\ -1]^t + \beta[0\ 1\ 3]^t : \alpha, \beta \in \mathbb{R}\}$ and let $b = A_{\star 1} + 2A_{\star 2} + A_{\star 3}$. Determine $\mathrm{Sol}(A, b)$.

23. Determine, if possible, the values of a, b so that the following systems have (i) no solutions, (ii) unique solution, and (iii) infinitely many solutions:

    (a) $x_1 + 2x_2 + x_3 = 1,\ -x_1 + 4x_2 + 3x_3 = 2,\ 2x_1 - 2x_2 + ax_3 = 3$

    (b) $x_1 + x_2 + 3x_3 = 2,\ x_1 + 2x_2 + 4x_3 = 3,\ x_1 + 3x_2 + ax_3 = b$

24. Show that the linear system $x + y + kz = 1,\ x - y - z = 2,\ 2x + y - 2z = 3$ has no solution for $k = 1$, and has a unique solution for each $k \neq 1$.